EXPERIMENT NO.9

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Aim: To implement a Private Ethereum Blockchain using Geth.

Theory:
Ethereum is an open-source, decentralized blockchain platform that enables developers to build and deploy smart contracts and decentralized applications (DApps). Unlike public blockchains such as the Ethereum mainnet, a Private Ethereum Blockchain is restricted to specific participants, making it ideal for development, testing, and enterprise applications where confidentiality, control, and performance are essential.
Geth (Go-Ethereum) is one of the most widely used Ethereum clients, written in the Go programming language. It allows users to run an Ethereum node, mine Ether, and interact with the blockchain through JSON-RPC, command-line, or JavaScript consoles.
Setting up a Private Ethereum Network involves creating a customized blockchain environment isolated from the public Ethereum network. It enables developers to test smart contracts, **consensus mechanisms,** and network behavior without incurring gas costs or depending on public validators.

Steps Involved:

1. Choosing a Network ID

Each private blockchain must have a unique network ID to differentiate it from public or other private networks.

2. Choosing a Consensus Algorithm

Ethereum supports different consensus mechanisms such as Proof of Work (PoW) and Proof of Authority (PoA). In private networks, Clique (PoA) is often used for faster block confirmation.

3. Creating a Genesis Block

The genesis block is the first block of the blockchain. It defines network parameters such as difficulty, gas limit, initial account balances, and consensus settings.

4. Initializing the Geth Database

The geth init command initializes the blockchain using the genesis.json file and prepares the data directory for node operation.

5. Setting up Networking

Nodes communicate using enodes (Ethereum Node URLs), which contain IP addresses and public keys for peer discovery and connection.

6. Running the Member Nodes

Each participant runs a Geth node that connects to the private network and participates in block validation and transaction processing.

7. Running a Signer (in Clique)

In the Clique consensus, designated signers validate and sign blocks. These accounts are pre-configured in the genesis file to ensure trusted block production.

Step 1: Install Geth

sudo apt update

```
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

sonam@Sonam:~$ sudo apt update
[sudo] password for sonam:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1578 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [252 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.7 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1167 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [225 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [22.8 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2763 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [644 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:19 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [544 B]
Get:20 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [28.8 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [6980 B]
Get:22 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:23 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [396 B]
Get:24 http://archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
```

sudo apt install software-properties-common

```
71 packages can be upgraded. Run 'apt list --upgradable' to see them.
sonam@Sonam:~$ sudo apt install software-properties-common
[sudo] password for sonam:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  python3-software-properties
The following packages will be upgraded:
  python3-software-properties software-properties-common
2 upgraded, 0 newly installed, 0 to remove and 69 not upgraded.
Need to get 44.4 kB of archives.
After this operation, 0 B of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 software-properties-common all 0.99.49.4 [14.4 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 python3-software-properties all 0.99.49.4 [30.0 kB]
Fetched 44.4 kB in 2s (21.9 kB/s)
(Reading database ... 40805 files and directories currently installed.)
Preparing to unpack .../software-properties-common_0.99.49.4_all.deb ...
Unpacking software-properties-common (0.99.49.4) over (0.99.49.3) ...
Preparing to unpack .../python3-software-properties_0.99.49.4_all.deb ...
Unpacking python3-software-properties (0.99.49.4) over (0.99.49.3) ...
Setting up python3-software-properties (0.99.49.4) ...
Setting up software-properties-common (0.99.49.4) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
sonam@Sonam:~$
```

sudo add-apt-repository -y ppa:ethereum/ethereum

```
sonam@Sonam:~$ sudo add-apt-repository -y ppa:ethereum/ethereum
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu/
Suites: noble
Components: main
'
More info: https://launchpad.net/~ethereum/+archive/ubuntu/ethereum
Adding repository.

Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble InRelease [18.1 kB]
Get:5 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble/main amd64 Packages [2584 B]
Get:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:7 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble/main Translation-en [772 B]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.2 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:16 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7380 B]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [13.2 kB]
```

sudo apt update

```
sonam@Sonam:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:5 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
69 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

sudo apt install geth

```
sonam@Sonam:~$ sudo apt install geth
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  geth
0 upgraded, 1 newly installed, 0 to remove and 69 not upgraded.
Need to get 16.6 MB of archives.
After this operation, 53.4 MB of additional disk space will be used.
Get:1 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble/main amd64 geth amd64 1.17.2+build8480+noble [16.6
 MB]
Fetched 16.6 MB in 44s (377 kB/s)
Selecting previously unselected package geth.
(Reading database ... 40805 files and directories currently installed.)
Preparing to unpack .../geth_1.17.2+build8480+noble_amd64.deb ...
Unpacking geth (1.17.2+build8480+noble) ...
Setting up geth (1.17.2+build8480+noble) ...
```

Step 2: Check the version of Geth on the Terminal

geth version

```
sonam@Sonam:~$ geth version
Geth
Version: 1.17.2-stable
Git Commit: be4dc0c4be2fe316dbdd0a73e48421f64978232f
Architecture: amd64
Go Version: go1.25.7
Operating System: linux
GOPATH=
GOROOT=
```

Step - 3: Create a Private Ethereum Network

1.Create a folder named, private_ethereum_setup

```
sonam@Sonam:~$ mkdir private_ethereum_setup
cd private_ethereum_setup
sonam@Sonam:~/private_ethereum_setup$
```

2.Create 2 subfolders named node1 and node2 in the folder private_ethereum_setup

```
sonam@Sonam:~$ mkdir private_ethereum_setup
cd private_ethereum_setup
sonam@Sonam:~/private_ethereum_setup$ mkdir node1 node2
sonam@Sonam:~/private_ethereum_setup$
```

3. Create accounts of two nodes

geth --datadir node1 account new

```
sonam@Sonam:~/private_ethereum_setup$ geth --datadir node1 account new
INFO [04-08|15:07:54.668] Maximum peer count                       ETH=50 total=50
INFO [04-08|15:07:54.670] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or dire
ctory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0x79F01BdF06B2995b5e70598BD197ed8446AD30d5
Path of the secret key file: node1/keystore/UTC--2026-04-08T15-08-03.180685050Z--79f01bdf06b2995b5e70598bd197ed8446ad30d
5

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!
```

geth --datadir node2 account new

```
sonam@Sonam:~/private_ethereum_setup$ geth --datadir node2 account new
INFO [04-08|15:14:33.247] Maximum peer count                       ETH=50 total=50
INFO [04-08|15:14:33.248] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or dire
ctory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0xa4181d0706124fdBa91a1ce105bc4284F3C0d448
Path of the secret key file: node2/keystore/UTC--2026-04-08T15-14-43.638226935Z--a4181d0706124fdba91a1ce105bc4284f3c0d44
8

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!
```

4. Create a genesis.json file in the folder, private_ethereum_setup nano genesis.json

```
sonam@Sonam: ~/private_etl

  GNU nano 7.2                                genesis.json *
{
  "config": {
    "chainId": 1234,
    "homesteadBlock": 0,
    "eip150Block": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "constantinopleBlock": 0,
    "petersburgBlock": 0,
    "istanbulBlock": 0,
    "ethash": {}
  },
  "nonce": "0x0",
  "timestamp": "0x5f5a0a92",
  "extraData": "0x0000000000000000000000000000000000000000000000000000000000000000",
  "gasLimit": "0x8000000",
  "difficulty": "0x400",
  "mixhash": "0x0000000000000000000000000000000000000000000000000000000000000000",
  "coinbase": "0x0000000000000000000000000000000000000000",
  "alloc": {
    "0x79F01BdF06B2995b5e70598BD197ed8446AD30d5": {
      "balance": "100000000000000000000"
    },
    "0xa4181d0706124fdBa91a1ce105bc4284F3C0d448": {
      "balance": "100000000000000000000"

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```

5.Initialize the nodes with the genesis file

```
sonam@Sonam:~/private_ethereum_setup$ geth --datadir node1 init genesis.json
INFO [04-08|15:29:42.126] Maximum peer count                       ETH=50 total=50
INFO [04-08|15:29:42.128] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or dire
ctory"
WARN [04-08|15:29:42.149] Sanitizing cache to Go's GC limits       provided=4096 updated=1254
INFO [04-08|15:29:42.155] Set global gas cap                       cap=50,000,000
INFO [04-08|15:29:42.159] Initializing the KZG library             backend=gokzg
INFO [04-08|15:29:42.161] Defaulting to pebble as the backing database
INFO [04-08|15:29:42.162] Allocated cache and file handles         database=/home/sonam/private_ethereum_setup/node1/get
h/chaindata cache=627.00MiB handles=524,288
INFO [04-08|15:29:42.300] Opened ancient database                  database=/home/sonam/private_ethereum_setup/node1/get
h/chaindata/ancient/chain readonly=false
INFO [04-08|15:29:42.301] Opened Era store                         datadir=/home/sonam/private_ethereum_setup/node1/geth
/chaindata/ancient/chain/era
INFO [04-08|15:29:42.303] State schema set to default              scheme=path
WARN [04-08|15:29:42.303] Head block is not reachable
INFO [04-08|15:29:42.305] Load database journal from disk
INFO [04-08|15:29:42.334] Opened ancient database                  database=/home/sonam/private_ethereum_setup/node1/get
h/chaindata/ancient/state readonly=false
INFO [04-08|15:29:42.334] State snapshot generator is not found
INFO [04-08|15:29:42.335] Starting snapshot generation             root=56e81f..63b421 accounts=0 slots=0 storage=0.00B
dangling=0 elapsed="60.868µs"
INFO [04-08|15:29:42.336] Resuming snapshot generation             root=56e81f..63b421 accounts=0 slots=0 storage=0.00B
dangling=0 elapsed=1.135ms
INFO [04-08|15:29:42.336] Initialized path database                triecache=16.00MiB statecache=16.00MiB buffer=64.00Mi
B state-history="last 90000 blocks" journal-dir=/home/sonam/private_ethereum_setup/node1/geth/triedb
INFO [04-08|15:29:42.338] Writing custom genesis block
INFO [04-08|15:29:42.339] Generated snapshot                       accounts=0 slots=0 storage=0.00B dangling=0 elapsed=4
.658ms
INFO [04-08|15:29:42.349] Successfully wrote genesis state         database=chaindata hash=b1453d..68041e
```

```
sonam@Sonam:~/private_ethereum_setup$ geth --datadir node2 init genesis.json
INFO [04-08|15:30:41.711] Maximum peer count                       ETH=50 total=50
INFO [04-08|15:30:41.713] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
WARN [04-08|15:30:41.721] Sanitizing cache to Go's GC limits       provided=4096 updated=1254
INFO [04-08|15:30:41.724] Set global gas cap                       cap=50,000,000
INFO [04-08|15:30:41.724] Initializing the KZG library             backend=gokzg
INFO [04-08|15:30:41.726] Defaulting to pebble as the backing database
INFO [04-08|15:30:41.726] Allocated cache and file handles         database=/home/sonam/private_ethereum_setup/node2/geth/chaindata cache=627.00
MiB handles=524,288
INFO [04-08|15:30:41.830] Opened ancient database                  database=/home/sonam/private_ethereum_setup/node2/geth/chaindata/ancient/chai
n readonly=false
INFO [04-08|15:30:41.830] Opened Era store                         datadir=/home/sonam/private_ethereum_setup/node2/geth/chaindata/ancient/chain
/era
INFO [04-08|15:30:41.830] State schema set to default              scheme=path
WARN [04-08|15:30:41.830] Head block is not reachable
INFO [04-08|15:30:41.830] Load database journal from disk
INFO [04-08|15:30:41.868] Opened ancient database                  database=/home/sonam/private_ethereum_setup/node2/geth/chaindata/ancient/stat
e readonly=false
INFO [04-08|15:30:41.869] State snapshot generator is not found
INFO [04-08|15:30:41.869] Starting snapshot generation             root=56e81f..63b421 accounts=0 slots=0 storage=0.00B dangling=0 elapsed="7.64
5µs"
INFO [04-08|15:30:41.869] Initialized path database                triecache=16.00MiB statecache=16.00MiB buffer=64.00MiB state-history="last 90
000 blocks" journal-dir=/home/sonam/private_ethereum_setup/node2/geth/triedb
INFO [04-08|15:30:41.869] Writing custom genesis block
INFO [04-08|15:30:41.869] Resuming snapshot generation             root=56e81f..63b421 accounts=0 slots=0 storage=0.00B dangling=0 elapsed="274.
232µs"
INFO [04-08|15:30:41.870] Generated snapshot                       accounts=0 slots=0 storage=0.00B dangling=0 elapsed=1.465ms
INFO [04-08|15:30:41.871] Successfully wrote genesis state         database=chaindata hash=b1453d..68041e
sonam@Sonam:~/private_ethereum_setup$
```

6.For configuring the boot node

```
sonam@Sonam:~/private_ethereum_setup$ sudo apt-get install bootnode
[sudo] password for sonam:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  bootnode
0 upgraded, 1 newly installed, 0 to remove and 69 not upgraded.
Need to get 6955 kB of archives.
After this operation, 21.9 MB of additional disk space will be used.
Get:1 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble/main amd64 bootnode amd64 1.14.13+build30700+noble [6955 kB]
Fetched 6955 kB in 21s (325 kB/s)
Selecting previously unselected package bootnode.
(Reading database ... 40812 files and directories currently installed.)
Preparing to unpack .../bootnode_1.14.13+build30700+noble_amd64.deb ...
Unpacking bootnode (1.14.13+build30700+noble) ...
Setting up bootnode (1.14.13+build30700+noble) ...
```

```
sonam@Sonam:~/private_ethereum_setup$ bootnode -nodekey boot.key -addr 127.0.0.1:30301
enode://a3facd52b0c0addde24983a6f9dce40348b465ab8aa39833140f68884b0b86e071058e5647daaac1ea78a606505f77891ab40f0c0514194bd008a07acab366ee@127.0.0
.1:0?discport=30301
Note: you're using cmd/bootnode, a developer tool.
We recommend using a regular node as bootstrap node for production deployments.
INFO [04-08|15:41:17.391] New local node record                    seq=1,775,662,877,389 id=0a7e4a359a00bf5f ip="invalid IP" udp=0 tcp=0
^C
sonam@Sonam:~/private_ethereum_setup$
```

Step - 4: Establish a Peer-Peer Connection between the nodes along with the bootnode

1. On the first Terminal, Use boot.key to run the boot node

```
sonam@Sonam:~/private_ethereum_setup$ bootnode -nodekey boot.key -addr 127.0.0.1:30301
enode://a3facd52b0c0addde24983a6f9dce40348b465ab8aa39833140f68884b0b86e071058e5647daaac1ea78a606505f77891ab40f0c0514194bd008a07acab366ee@127.0.0
.1:0?discport=30301
Note: you're using cmd/bootnode, a developer tool.
We recommend using a regular node as bootstrap node for production deployments.
INFO [04-08|15:41:17.391] New local node record                    seq=1,775,662,877,389 id=0a7e4a359a00bf5f ip="invalid IP" udp=0 tcp=0
^C
sonam@Sonam:~/private_ethereum_setup$ bootnode -nodekey boot.key -addr 127.0.0.1:30301
enode://a3facd52b0c0addde24983a6f9dce40348b465ab8aa39833140f68884b0b86e071058e5647daaac1ea78a606505f77891ab40f0c0514194bd008a07acab366ee@127.0.0
.1:0?discport=30301
Note: you're using cmd/bootnode, a developer tool.
We recommend using a regular node as bootstrap node for production deployments.
INFO [04-08|15:46:15.574] New local node record                    seq=1,775,663,175,573 id=0a7e4a359a00bf5f ip="invalid IP" udp=0 tcp=0
```

On the second Terminal, Run Node 1

```
sonam@Sonam:~$ cd ~/private_ethereum_setup
sonam@Sonam:~/private_ethereum_setup$ geth --datadir node1 --networkid 1234 --port 30303 --bootnodes "enode://f04c7fee38c1d653077101cb45ff520ecbb42aec1fbcaa
229fe4a936285d05b5a546d9624edd235605d63755cfaa27a2b1c277e25bc31f924145f908edf847fb@127.0.0.1:0?discport=30301" --unlock 0xF806583aD18ee14C561586c6D3F1467E04
4BC60b --miner.etherbase 0xF806583aD18ee14C561586c6D3F1467E044BC60b --mine --allow-insecure-unlock console
INFO [04-08|17:16:00.202] Maximum peer count                       ETH=50 total=50
INFO [04-08|17:16:00.203] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-08|17:16:00.208] Set global gas cap                       cap=50,000,000
INFO [04-08|17:16:00.208] Initializing the KZG library             backend=gokzg
INFO [04-08|17:16:00.257] Allocated trie memory caches             clean=154.00MiB dirty=256.00MiB
INFO [04-08|17:16:00.258] Using pebble as the backing database
INFO [04-08|17:16:00.258] Allocated cache and file handles         database=/home/sonam/private_ethereum_setup/node1/geth/chaindata cache=512.00MiB handles=
524,288
INFO [04-08|17:16:00.283] Opened ancient database                  database=/home/sonam/private_ethereum_setup/node1/geth/chaindata/ancient/chain readonly=f
alse
INFO [04-08|17:16:00.283] State scheme set to already existing     scheme=hash
INFO [04-08|17:16:00.285] Initialising Ethereum protocol           network=1234 dbversion=8
INFO [04-08|17:16:00.285]
INFO [04-08|17:16:00.285] ----------------------------------------------------------------------------------------------------------------------------
----------------------
INFO [04-08|17:16:00.285] Chain ID:  1234 (unknown)
INFO [04-08|17:16:00.285] Consensus: Clique (proof-of-authority)
INFO [04-08|17:16:00.285]
INFO [04-08|17:16:00.285] Pre-Merge hard forks (block based):
INFO [04-08|17:16:00.285]  - Homestead:                    #0       (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrad
es/homestead.md)
INFO [04-08|17:16:00.285]  - Tangerine Whistle (EIP 150): #0       (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrad
es/tangerine-whistle.md)
INFO [04-08|17:16:00.285]  - Spurious Dragon/1 (EIP 155): #0       (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrad
es/spurious-dragon.md)
INFO [04-08|17:16:00.285]  - Spurious Dragon/2 (EIP 158): #0       (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrad
es/spurious-dragon.md)
INFO [04-08|17:16:00.285]  - Byzantium:                    #<nil> (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/
byzantium.md)
INFO [04-08|17:16:00.285]  - Constantinople:               #<nil> (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/
constantinople.md)
INFO [04-08|17:16:00.285]  - Petersburg:                   #<nil> (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/
petersburg.md)
```

On the third Terminal, Run Node 2

```
sonam@Sonam:~/private_ethereum_setup$ geth --datadir node2 \
--networkid 1234 \
--port 30304 \
--authrpc.port 8552 \
--bootnodes "enode://f04c7fee38c1d653077101cb45ff520ecbb42aec1fbcaa229fe4a936285d05b5a546d9624edd235605d63755cfaa27a2b1c277e25bc31f924145f908edf847fb@127.0.
0.1:0?discport=30301" \
console
INFO [04-08|17:19:46.610] Maximum peer count                       ETH=50 total=50
INFO [04-08|17:19:46.611] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-08|17:19:46.615] Set global gas cap                       cap=50,000,000
INFO [04-08|17:19:46.615] Initializing the KZG library             backend=gokzg
INFO [04-08|17:19:46.674] Allocated trie memory caches             clean=154.00MiB dirty=256.00MiB
INFO [04-08|17:19:46.674] Using pebble as the backing database
INFO [04-08|17:19:46.674] Allocated cache and file handles         database=/home/sonam/private_ethereum_setup/node2/geth/chaindata cache=512.00MiB handles=
524,288
INFO [04-08|17:19:46.693] Opened ancient database                  database=/home/sonam/private_ethereum_setup/node2/geth/chaindata/ancient/chain readonly=f
alse
INFO [04-08|17:19:46.693] State scheme set to already existing     scheme=hash
INFO [04-08|17:19:46.694] Initialising Ethereum protocol           network=1234 dbversion=8
INFO [04-08|17:19:46.694]
INFO [04-08|17:19:46.694] ----------------------------------------------------------------------------------------------------------------------------
----------------------
INFO [04-08|17:19:46.694] Chain ID:  1234 (unknown)
INFO [04-08|17:19:46.694] Consensus: Clique (proof-of-authority)
INFO [04-08|17:19:46.694]
INFO [04-08|17:19:46.694] Pre-Merge hard forks (block based):
INFO [04-08|17:19:46.694]  - Homestead:                    #0       (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrad
es/homestead.md)
INFO [04-08|17:19:46.694]  - Tangerine Whistle (EIP 150): #0       (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrad
es/tangerine-whistle.md)
INFO [04-08|17:19:46.694]  - Spurious Dragon/1 (EIP 155): #0       (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrad
es/spurious-dragon.md)
INFO [04-08|17:19:46.694]  - Spurious Dragon/2 (EIP 158): #0       (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrad
es/spurious-dragon.md)
INFO [04-08|17:19:46.694]  - Byzantium:                    #<nil> (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/
byzantium.md)
INFO [04-08|17:19:46.694]  - Constantinople:               #<nil> (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/
constantinople.md)
INFO [04-08|17:19:46.694]  - Petersburg:                   #<nil> (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/
```

Step - 5 : Exploring the network by attaching JavaScript console to Node 1 1. Fetch network status

1. Fetch Network Stat

```
> net
{
  listening: true,
  peerCount: 1,
  version: "1234",
  getListening: function(callback),
  getPeerCount: function(callback),
  getVersion: function(callback)
}
```

2. To Fetch the Number of Blocks Mined

```
> web3.eth.blockNumber
87
>
```

3. To check the balances of the account

```
> eth.getBalance(eth.accounts[0])
100000000000000000000
> INFO [04-08|17:24:06.447] Successfully sealed new block           number=98 sealhash=2be448..3b41b1 hash=e638a
INFO [04-08|17:24:06.447] Commit new sealing work                 number=99 sealhash=d3604f..c254bb txs=0 gas=0
```

4. To Fetch the Details of the Latest Mined Block

```
> eth.getBlock("latest")
{
  difficulty: 2,
  extraData: "0xd883010d0e846765746888676f312e32312e36856c696e75780000000000000000c78bebc5a700db7175113eb767fa794d1cb80f6c66d0fafe227a3c976fec2ab644df6
b049691f53fcccce5e98d9e8c4d6c15392a0d4138cbd1e6289a200",
  gasLimit: 9425278,
  gasUsed: 0,
  hash: "0x3ffdc7b4b744511412d46e348c4fb3501e575cfff46a666e28287cbf5fbd39d7",
  logsBloom: "0x00000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000
```

```
> eth.getBlock(5)
{
  difficulty: 2,
  extraData: "0xd883010d0e846765746888676f312e32312e36856c696e757800000000000000009cd0aba78569c63072e48bf7d74951971cd439f143b2ba1ec16796208121fad57e2555696fb
193456c2b564d512deaf478de9b92ae24ad2d52a7641b987c5fb00",
  gasLimit: 8039132,
  gasUsed: 0,
  hash: "0x53e15f36b6e1d6f8e2c8416e69ea94f8545e3f035242d23c33734c9694d03aab",
  logsBloom: "0x00000000000000
```

```
> eth.sendTransaction({
......   from: "0xF806583aD18ee14C561586c6D3F1467E044BC60b",
......   to: "0xE951476352616Eb7eeaf1D7ca1701A95AFD4EF5b",
......   value: web3.toWei(5, "ether")
...... })INFO [04-08|17:33:24.494] Looking for peers                    peercount=1 tried=0 static=0

INFO [04-08|17:33:24.828] Setting new local account              address=0xF806583aD18ee14C561586c6D3F1467E044BC60b
INFO [04-08|17:33:24.829] Submitted transaction                  hash=0x6280beb0f83a01b609ca7b5d6fa5d7e4956abedc038686e49f2c4114e050af66 from=0xF806583aD1
8ee14C561586c6D3F1467E044BC60b nonce=0 recipient=0xE951476352616Eb7eeaf1D7ca1701A95AFD4EF5b value=5,000,000,000,000,000,000
"0x6280beb0f83a01b609ca7b5d6fa5d7e4956abedc038686e49f2c4114e050af66"
> INFO [04-08|17:33:26.013] Commit new sealing work                number=210 sealhash=5aea70..d0a89f txs=1 gas=21000 fees=2.1e-05 elapsed=3.564ms
INFO [04-08|17:33:27.018] Successfully sealed new block          number=210 sealhash=5aea70..d0a89f hash=6ca85e..44851f elapsed=1.005s
INFO [04-08|17:33:27.019] Commit new sealing work                number=211 sealhash=9c2fe2..a82c63 txs=0 gas=0     fees=0      elapsed="572.686µs"
```

9.Check the Balances of Sender and Receiver

```
> eth.getBalance("0xF806583aD18ee14C561586c6D3F1467E044BC60b")
95000000000000000000
> INFO [04-08|17:33:54.063] Looking for peers                    peercount=1 tried=0 static=
```

```
> eth.getBalance("0xE951476352616Eb7eeaf1D7ca1701A95AFD4EF5b")
105000000000000000000
> INFO [04-08|17:34:12.015] Successfully sealed new block          number=219 sealhash=b37ab9..1705
INFO [04-08|17:34:12.015] Commit new sealing work                number=220 sealhash=45f46e..052db6
```

10.Check the Details of the Transaction on Node 1 Terminal

```
> eth.getTransaction("0x6280beb0f83a01b609ca7b5d6fa5d7e4956abedc038686e49f2c4114e050af66")
{
  blockHash: "0x6ca85ed8156677569c9623dbb867dee1acec022443bcc9699e0582d16e44851f",
  blockNumber: 210,
  chainId: "0x4d2",
  from: "0xf806583ad18ee14c561586c6d3f1467e044bc60b",
  gas: 21000,
  gasPrice: 1000000000,
  hash: "0x6280beb0f83a01b609ca7b5d6fa5d7e4956abedc038686e49f2c4114e050af66",
  input: "0x",
  nonce: 0,
  r: "0xb4c908f5eb7e9239bcef728a0e62792c950767c21e2c8968f5ec06376a71dd71",
  s: "0x43f62dc4ab8baba7bdcf4d63c081158154ae0310108a1c28da93d43f0244b51a",
  to: "0xe951476352616eb7eeaf1d7ca1701a95afd4ef5b",
  transactionIndex: 0,
  type: "0x0",
  v: "0x9c8",
  value: 5000000000000000000
}
> INFO [04-08|17:37:27.012] Successfully sealed new block          number=258 sealhash=e7e98d..d3d85c hash=8f78cc..23ca47 elapsed=5.003s
```

11. To check the contents in the Mempool - Transaction Pool

```
INFO [04-08|17:38:27.008] Commit new sealing work                  number=271 sealhash=c3f126..0b165b txs=0 gas=0     fees=0
> txpool.contentINFO [04-08|17:38:30.759] Looking for peers                      peercount=1 tried=0 static=0

{
  pending: {},
  queued: {}
}
> INFO [04-08|17:38:32.008] Successfully sealed new block           number=271 sealhash=c3f126..0b165b hash=0ed475..194b30 ela
INFO [04-08|17:38:32.008] Commit new sealing work                 number=272 sealhash=769bac..4d5ff6 txs=0 gas=0     fees=0
INFO [04-08|17:38:37.015] Successfully sealed new block           number=272 sealhash=769bac..4d5ff6 hash=5bfe94..bbb14b elaps
```

12. To check the contents in the Mempool - Transaction Pool

```
INFO [04-08|17:40:02.010] Commit new sealing work                  number=290 sealhash=524bdc..90051
> txpool.status
{
  pending: 0,
  queued: 0
}
```

Conclusion:-

The experiment successfully implemented a Private Ethereum Blockchain using Geth and Clique (PoA) consensus. By initializing a custom genesis block, an isolated network was established for zero-cost testing. This setup proved that authorized signers can effectively validate transactions without the computational overhead of public networks, providing a secure environment for enterprise DApp development.